OR Final Project - Optimizing Bike Lanes in Taipei

Group C

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Introduction

Taipei boasts a well-established YouBike system, widely used by both locals and tourists for commuting, last-mile transit, and leisure. However, biking in the city can often be unsafe and inconvenient due to limited and inconsistent bike lane infrastructure.

As public interest in sustainability and green urban mobility grows, Taipei's city government has begun investing in the expansion of its bike lane network. Yet, the key challenge lies in where to build. With limited construction budgets, it is no longer sufficient to simply add more kilometers of lanes. New infrastructure must be thoughtfully connected, strategically placed, and informed by actual usage data to maximize public benefit and impact.

The link to our repository is here.

Problem Definition

For our project, we aim to identify the road segments that would yield the greatest benefit from the construction of bike lanes, as well as to determine the appropriate level of infrastructure to implement. The objective is to maximize the overall utility for cyclists throughout the city.

On the benefit side, cycling demand serves as a key indicator of the potential utility derived from bike lane investments. The demand modeling approach employed in this project is detailed in the Data Processing section. In addition, spatial continuity plays an important role: when adjacent road segments are both equipped with bike lanes, the resulting consistency in the riding experience contributes positively to overall utility. Moreover, for road segments with higher safety risks, constructing more advanced infrastructure, such as physically separated bike lanes or improved pavement, is expected to yield greater benefits.

Our optimization formulation incorporates two primary constraints. Firstly, the total construction cost for the proposed bike lane network must remain strictly within the confines of the annual budget. Secondly, to ensure the resulting bike lane network is genuinely continuous and not merely a star-like configuration radiating from isolated intersection points, we impose a constraint on the number of activated intersections for each road segment. This latter constraint is crucial for promoting effective connectivity and usability across the network.

Mathematical Formulation

Decision Variables

Let R denote the set of all candidate road segments in Taipei City for potential bike lane construction. Define the set of bike lane infrastructure levels as $G = 1, 2$, where level 1 corresponds to basic infrastructure, such as painted lane markings indicating bike lanes, and level 2 represents enhanced infrastructure, including physically separated bike lanes with upgraded pavement and safety features distinguishing them from vehicular traffic lanes.

Our core decision variable, x_{ik}, is defined as follows:

$$x_{ik} = \begin{cases} 1, & \text{if road i is selected to build a bike with level k} \\ 0, & \text{otherwise} \end{cases} \quad , i \in R, k \in G$$

An auxiliary decision variable is introduced to later linearize the objective function. We define y_{ij} as the following:

$$y_{ij} := \begin{cases} 1, & \text{if the intersection of roads } i \text{ and } j \text{ is served,} \\ 0, & \text{otherwise} \end{cases} \quad , (i,j) \in I$$

An intersection (i, j) is said to be "served" if both road i and j have bike lanes built on. We let I denote the set of all intersections (i, j), which we elaborate in the Adjacency Matrix section.

Adjacency Matrix

To compute the utility of two adjacent bike lanes, we define an adjacency matrix which indicates whether a road pair, i and j, are connected. Let A be our adjacency matrix:

$$[A_{ij}] := \begin{cases} 1, & \text{if road } i \text{ and road } j \text{ are adjacent,} \\ 0, & \text{otherwise} \end{cases} \quad , i, j \in R$$

We can further define a set of road intersections as follows:

$$I = \{(i,j) | A_{ij} = 1; i, j \in R, i < j\},$$

where $i < j$ ensures that no intersections are counted twice.

Objective Function

We design our objective function as the sum of two kinds of utility: the utility directly associated with building one specific bike lane, and the utility of bike lane consistency. Mathematically, the first part is the sum of all utility of road, and the second part is the sum of all utility of intersections of bike lanes pair, which could be formulated as follows: [1]

$$f(x_{ik}) = \mu \sum_{i \in R} U_i^{\text{road}} \cdot (x_{i1} + p_i x_{i2}) + (1 - \mu) \sum_{(i,j) \in I} U_{ij}^{\text{int}} (x_{i1} + x_{i2})(x_{j1} + x_{j2}),$$

where U_i^{road} denotes the utility of road i if there is a bike lane built on it; p_i as the relative utility advantage of buliding level 2 bike lane than level 1 on road i, which we, for the sake of simplicity, set to be proportional to the road danger, giving incentive to upgrading more dangerous roads to more protected bike lanes. U_{ij}^{road} is the utility at the intersection (i, j), and μ as the relative importance of road utility over intersection utility, $\mu \in [0, 1]$. With the auxiliary decision variable y_{ij}, the objective function is accordingly linearized as follows:

$$f^L(x_{ik}, y_{ij}) = \underbrace{\mu \sum_{i \in R} U_i^{\text{road}} \cdot (x_{i1} + p_i x_{i2})}_{\text{Utility of Roads}} + \underbrace{(1 - \mu) \sum_{(i,j) \in I} U_{ij}^{\text{int}} y_{ij}}_{\text{Utility of Intersections}}$$

Utility of Roads The utility contributed by an individual road segment, expressed as $U_i^{\text{road}} \cdot (x_{i1} + p_i x_{i2})$, can be decomposed into two components: the intrinsic or potential utility U_i^{road}, and the realized infrastructure decision, represented by $x_{i1} + p_i x_{i2}$.

The potential utility U_i^{road} is defined as a Cobb Douglas function of two indicators: cycling demand and road length. Let D_i^{norm} denote the normalized cycling demand and L_i^{norm} the normalized length of

[1] Note the similarity between this objective function and the Hamiltonian of an Ising model, with which comparisons could serve as a starting point into some deeper theoretical considerations (Phase transitions, Symmetry(-breaking), ...), or heuristic algorithms inspired by the Metropolis-Hastings method often used to minimize this Hamiltonian.
Additionally, the objective function could also be seen analogous to the first two terms of a many body expansion, and could naturally be expanded to higher order terms, not only considering adjacent roads but also next-nearest roads and so on, to better control connectivity of the road network.

road segment i. Note that both parameters are normalized to ensure comparability, and shifted by 3 to guarantee non-negativity.

The parameter $\alpha \in [0, 1]$ is a tunable hyperparameter that controls the relative importance of road length versus cycling demand in the utility calculation. Then, U_i^{road} is defined as follows:

$$U_i^{\text{road}} = (L_i^{\text{norm}})^{\alpha}(D_i^{norm})^{1-\alpha}, i \in R$$

Longer road segments intrinsically generate greater utility when equipped with bike lanes, as they provide extended coverage for cyclists. Similarly, road segments with higher cycling demand contribute more significantly to overall utility, reflecting their greater usage potential.

Individual intersection utility U_{ij}^{int}: The potential utility of an intersection (i, j) is set to be the demand of the intersection D_{ij}^{norm}, discussed in detail later:

$$U_{ij}^{\text{int}} = D_{ij}^{\text{int}}$$

Constraints

We consider several constraints in the model: the total construction budget, constraints linking our decision variables, and constraints controlling the topology of the resulting network.

Construction Cost: Let C_1 denote the unit construction cost of a level 1 bike lane and C_2 that of a level 2 bike lane. Let L_i represent the length of road segment i. The total construction cost is constrained by a budget limit B. The construction cost budget constraint could be formulated as follows:

$$\sum_{i \in R}(C_1 x_{i1} + C_2 x_{i2}) \cdot L_i \leq B$$

Due to the unavailability of detailed construction cost data, we assume a simplified cost structure in which C_2 is scaled from C_1 by a constant factor w, i.e., $C_2 = w \cdot C_1$. We further define B^L as the total budget expressed in terms of road length (in meters), where $B^L = \frac{B}{C_1}$. By dividing both sides of the original constraint by C_1, the budget constraint can be reformulated as follows:

$$\sum_{i \in R}(x_{i1} + w \cdot x_{i2}) \cdot L_i \leq B^L$$

This transformation allows the budget constraint to be expressed in terms of the actual length of roads, which provides a more intuitive interpretation. It also offers a practical modeling simplification in the absence of detailed real-world budget data, enabling analysis without requiring precise cost parameters. For most scenarios that we experiment on, the constant factor w is set to 3.

Linking of x and y: The utility of intersection (i, j) is realized only when $y_{ij} = 1$, which indicates that both road segments in the adjacent pair $(i, j) \in I$ are equipped with bike lanes. Therefore, the relationship between y_{ij} and the individual decisions x_i and x_j can be represented as follows:

$$\begin{cases} y_{ij} \leq x_{i1} + x_{i2} \\ y_{ij} \leq x_{j1} + x_{j2} \end{cases} \quad \forall (i, j) \in I$$

Both of these constraints are necessary, due to our definition of I; if $(i, j) \in I$, (j, i) is not. Additionally, we must also enforce that at most one upgrade level is selected per road:

$$x_{i1} + x_{i2} \leq 1 \quad \forall i \in I$$

Maximum Degree Constraint for x_{ik}: If we consider the formulation so far, there is still one unaddressed issue: The solver will tend to maximize y_{ij}'s. Considering the case of two adjacent roads having been selected, the optimal way to place a third road would most likely also be adjacent to those two. That way, two more y_{ij}'s could be set to one, for the construction cost of only one road. The same goes for adding yet another, fourth road; placing it adjacent to the other three roads would be the most cost-efficient way of maximizing the objective function. In the end this will cause solutions to have a lot

of roads built adjacent to the same few intersections, and a bunching up of planned bike lanes instead of a network with longer continuous bike lanes.
There are a few ways this behavior could be suppressed[2], a straightforward option is to limit the number of roads that can intersect in one intersection. If this limit is set, a branching and bunching of bike lanes are discouraged, but connectedness is still possible:

$$\sum_{(i,k)\in I} y_{ik} + \sum_{(k,j)\in I} y_{kj} \leq d \quad \forall k \in R$$

Essentially, this forces only *a maximum of* d of the y_{ij} to be able to be set to one for each intersection; and since y_{ij} describes *two* intersecting roads, this imposes precisely the restriction discussed above. The term d is the hyperparameter of the maximum degree, where $d \in \mathbb{N}$.

Complete Formulation

In sum, the complete integer linear program could be formulated as follows:

$$\textbf{max} \quad \mu \sum_{i\in R} U_i^{\text{road}} \cdot (x_{i1} + p_i x_{i2}) + (1-\mu) \sum_{(i,j)\in I} U_{ij}^{\text{int}} y_{ij}$$

$$\textbf{s.t.:} \quad \sum_{i\in R} (x_{i1} + w \cdot x_{i2}) \cdot L_i \leq B^L \tag{1}$$

$$y_{ij} \leq x_{i1} + x_{i2} \qquad \forall (i,j) \in I \tag{2}$$

$$y_{ij} \leq x_{j1} + x_{j2} \qquad \forall (i,j) \in I \tag{3}$$

$$x_{i1} + x_{i2} \leq 1 \qquad \forall i \in R \tag{4}$$

$$\sum_{(i,k)\in I} y_{ik} + \sum_{(k,j)\in I} y_{kj} \leq d \qquad \forall k \in R \tag{5}$$

$$x_{ik} = \{0,1\} \qquad \forall i \in R, k \in G \tag{6}$$

$$y_{ij} = \{0,1\} \qquad \forall (i,j) \in I \tag{7}$$

$$\alpha, \mu \in [0,1] \tag{8}$$

$$B^L \in \mathbb{R}^+ \tag{9}$$

$$d \in \mathbb{N} \tag{10}$$

Real World Data

Demand Modeling

We use datasets of YouBike usage to build a simple demand model for cycling infrastructure in Taipei, described in more detail below.

Local demand - demand induced by one station: The demand, that is generated by one station, is estimated from YouBike trip data. A public dataset contains the number of trips traveled with YouBikes between any two stations in the Taipei YouBike network. From the calculated (euclidian) distance between these pairs of YouBike-Stations and the respective number of trips, the histogram in Fig. 1 (a) was calculated. The trip frequency as a function of distance is denoted by $f(r)$, where r is the traveling distance, and is very well described by a Log-normal-distribution.
We can interpret the cumulative density function $CDF(r)$ of that distribution, shown in Fig. 1 (b), as the proportion of people that have *stopped* cycling at a distance r. Then intuitively, $[1 - CDF(r)]$ will be the share of people that are still riding. In one dimension (imagine demand only along one straight road with no intersections), this is equivalent to the demand at that distance. This is plotted as the green line in Fig. 1 (c).

[2] one option could also be to discourage bike lanes being built close together via a suitable term in the objective function (perhaps inspired by longer range repulsive interaction terms from physics). This would encourage bike lanes to straighten out and softly discourage many bike lanes intersecting in one point. Disadvantage: computationally expensive (up to numberOfRoads^2 interaction terms in objective function), Advantage: could maybe control the behavior of the formed road network more finely via the exact form of the interaction

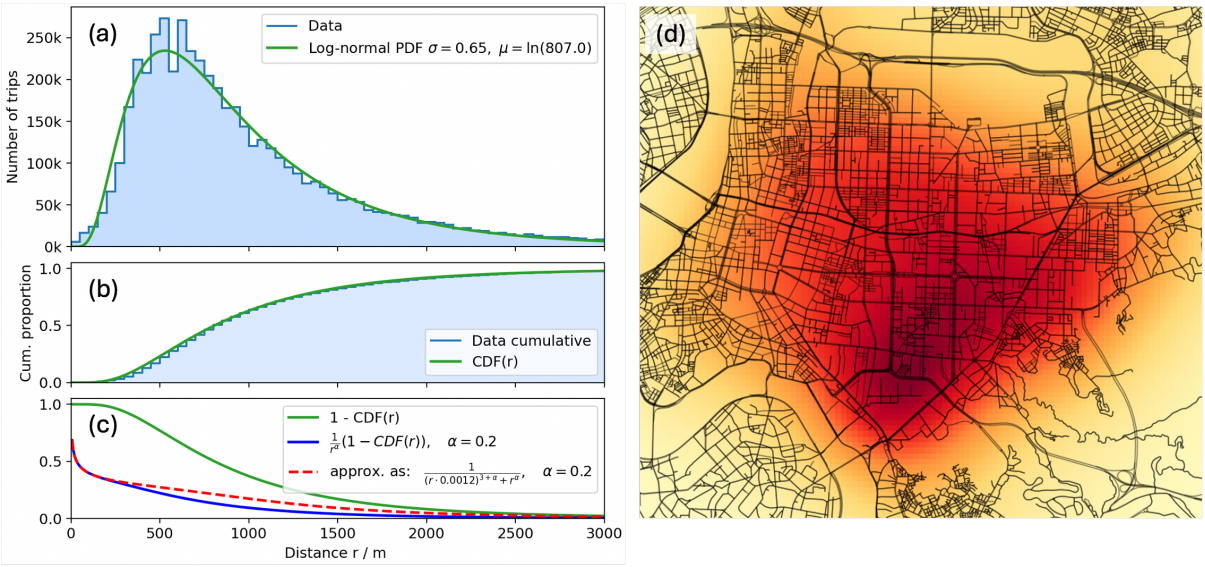


Figure 1: (a) shows a histogram of trip length frequency, (b) shows the respective cumulative density function. In (c), the demand corresponding to this, with an exemplary dimensionality exponent $\alpha = 0.2$, is shown. (d) shows the resulting demand distribution over Taipei (arbitrary units for the color scale, red corresponds to high demand).

The one dimensional case is of course unrealistic for our model, and would result in an overly slow decay of demand over distance, and a very uniform demand distribution over the length scale of the city. Therefore, we have to correct this demand kernel, by considering the "dimensionality" of the road network. We start by considering the other extreme case of cyclists driving away isotropically from their starting station in two dimensions. The demand will then be:[3]

$$D(r) = \frac{1 - CDF(r)}{r}$$

This will, on the other hand, result in unrealistically fast decay, since the assumption of isotropy is very inaccurate: Flow of cyclists away from stations is still constrained by the road network, so an assumed $1/r$ decay factor is exaggerated. In reality, we should expect to see some intermediate case, expressed with an exponent $\alpha \in [0, 1]$:

$$D_\alpha(r) = \frac{1 - CDF(r)}{r^\alpha} \tag{11}$$

For $\alpha = 0$ and $\alpha = 1$ this corresponds to the above-discussed cases of one and two-dimensional demand decay. It is not trivial to estimate the best value for this, and since it would be beyond the scope of this work, we chose an intermediate value of $\alpha = 0.2$.

To speed up later calculations, we can approximate the demand above with a function:

$$D_\alpha(r) = \frac{1}{(r \cdot 0.0012)^{3+\alpha} + r^\alpha}, \tag{12}$$

Shown as the red dashed line in Fig. 1 (c).

Global demand - sum of all demands: If we now want to model the demand $D(\vec{r})$ as a function of position $\vec{r}$ in all of Taipei, we can simply sum up the demands induced by individual stations:

$$D^{pos}(\vec{r}) = \sum_{i \in S^I} f_{0,i} \cdot D_\alpha(|\vec{r} - \vec{r_i}|), \tag{13}$$

Where S^I is the set of YouBike stations and $f_{0,i}$ is the total number of trips ending in station i, to account for the fact that busier stations will induce more demand around them; $D_\alpha(\cdot)$ is the local demand kernel discussed in the last paragraph.

[3] This follows intuitively because the demand at a distance r can be seen as the flow of cyclists flowing through a circle with that radius per unit circumference. And since the circumference scales proportional to r, the demand will scale with $1/r$.

We transform the Taipei City map into a grid system with a 50-meter resolution. For each grid cell, we calculated its demand using the function $D^{pos}(\vec{r})$, thereby generating a comprehensive demand map for the city. The result of this is shown in Fig. 1 (d).

Roads Data

Another core dataset of our research is the road data of Taipei city. Fortunately, Taipei City Hall provides an openly accessible dataset of road segments and their corresponding geometry information, width, length, and other attributes. We then perform further data processing techniques on this raw data to ensure it optimally fits the specific requirements of our analysis.

Demand of Roads Utilizing the geopandas and shapely modules, we proceeded from the demand modeling part to calculate the demand for each road segment. While it is intuitive to assign a road's demand based on its geometric position, a road is inherently represented as a LineString rather than a discrete point. Therefore, to accurately capture its spatial demand, we adopted a sampling approach: several points are randomly sampled along the LineString of each road. The demand for these sampled points is then retrieved using the demand function $D^{pos}(\cdot)$, and their average value is assigned as the overall demand for that specific road segment. Finally, we normalize the calculated road demand and shift by 3 (to ensure positive) to align it with the D^{norm} formulation used in our model.

Normalization of Length: In order to calculate the potential demand for each road, $U_i^{\text{road}} = (D_i^{\text{norm}})^{\alpha}(L_i^{\text{norm}})^{(1-\alpha)}$, we also need the normalized length. We implement the same trick as normalizing demand: normalize the length and shift by 3 to ensure positive. To calculate the potential utility for each road, $U_i^{\text{road}} = D_i^{\text{norm}} L_i^{\text{norm}}$, we require the normalized length L_i^{norm}. We apply the same normalization technique used for demand: the raw length values are normalized and then shifted by 3 to ensure all resulting values are positive. It's important to clarify that the original length data is retained and not replaced by the normalized length. This is because the unnormalized length is essential for calculating the construction cost.

Overlapping Roads Data with Existing Bike Lane Map: In our experiment, we conduct two distinct analyses: first, we evaluate our solution against the current bike lane policy, assuming no existing infrastructure. Then, we simulate a "clean slate" by eliminating current bike lanes to provide recommendations for new construction. For both scenarios, integrating existing bike lane data is crucial. We achieved this by merging publicly available Taipei Bike Lane data with our road network. Specifically, we added a binary attribute, `has_bike_lane`, to our road data; a road is flagged 1 if it overlaps with any existing bike lane segment.

Danger of Road (p_i): We intuitively approximate a road's degree of danger by the frequency of car accidents occurring in its proximity. Using the Taipei City Car Accident Data for 2024, which includes accident records with corresponding geometric information, we first generated a Taipei City danger heatmap. This was achieved by applying the same 50-meter grid-splitting method used in the Global Demand section, where the number of accidents is assigned to each grid cell.

To assign a danger degree to individual roads, we query the geometry of each road against this danger heatmap. Finally, the obtained danger values are normalized and then shifted by 3 to ensure all values are positive, aligning with our formulation's requirements.

Adjacency Matrix and Intersections

Leveraging road network data acquired from Taipei City Hall, our analysis directly generates intersection data (equivalent to the intersection set I), rather than explicitly constructing a traditional adjacency matrix A. This process primarily utilizes the geopandas and shapely Python modules to identify potential intersections between pairs of road segments (i, j).

As not all existing intersections were detected initially, we expanded the geometry objects of the road segments before finding their overlap, to ensure more robust intersection detection. The demand of an intersection D_{ij}, was then sampled from the demand grid computed above at its respective position, normalized to a mean of zero and std.-deviation of one, then shifted by three to obtain D_{ij}^{norm}.

Heuristic Algorithms

A Very Naive Heuristic Algorithm

To quickly identify promising bike lane locations, we can employ a naive greedy algorithm, denoted as Alg^N. This method operates on the simple, intuitive principle of always selecting the road that offers the highest immediate utility. The algorithm starts from identifying the road segment with the highest potential utility, with index denoted as max_id. Next, the algorithm defines a subset of intersection set I, where only road segments adjacent to road max_id are included, denoted as I'. Mathematically, $I' = \{(i,j)|i = \text{max_id} \vee j = \text{max_id}\}$ The algorithm then counts the number of intersections that would be activated if we build a bike lane on road max_id. If the count is larger than the maximum degree d, building bike lane on it would violate the maximum constraint, thus we eliminate it from the candidate roads and go through the next iteration. Also, if any partner of road max_id in the intersection set is activated and the maximum degree constraint of it is binding, we also remove max_id from our candidates. If the iteration passes all examinations above, we build bike lane on road max_id, then calculate the gain of road utility and intersection utility, and deduct the total budget length. Finally, if we decide to build bike lane on road max_id, and if the number of activated intersections that the road with max_id participates in is equal to the degree d, activation of any other roads adjacent to it would violate the maximum degree constraint. Therefore, we also remove these roads from candidates afterwards. While not necessarily optimal, this approach provides a rapid solution, conceptually mirroring the greedy strategy used in the traditional knapsack problem. Its step-by-step logic is presented in the pseudocode that follows.

---

procedure Alg^N(Road, Intersections)
  Initialize Numerical Vars: **roadUtils, intUtils**
  Initialize List Vars: **Type_1, Type_2, ActivatedInt, candidates**
  while $B^L > 0$ **do** ▷ $\mathcal{O}(|R|)$
    max_id = Roads["Ui"].idxmax() ▷ Find road with highest U_i: $\mathcal{O}(|R|)$
    IntSubset = Intersections["i" or "j" == max_id] ▷ Define the subset I': $\mathcal{O}(|I|)$
    if count(max_id's activated partner) $> d$ **then** ▷ $\mathcal{O}(|I|)$
      candidates.remove(max_id) and Continue
    if any(count(degree of max_id's activated partner)$\geq d \; \forall$ max_id's partner) **then** ▷ $\mathcal{O}(|I|)$
      candidates.remove(max_id) and Continue
    Initialize Boolean Var: **built = False** ▷ $\mathcal{O}(1)$
    if $p_{\text{max_id}} >$ thres **then** ▷ If dangerous enough, build type two
      Update(max_id, Type_2, ActivatedInt, Roads, Intersections) ▷ $\mathcal{O}(|I|)$
      candidates.remove(max_id) and built = True
    else ▷ Othewise, build type one
      Update(max_id, Type_1, ActivatedInt, Roads, Intersections) ▷ $\mathcal{O}(|I|)$
      candidates.remove(max_id) and built = True
    if built == True and road max_id is binding **then**
      Remove all adjacent partners from IntSubset from candidates ▷ $\mathcal{O}(|I|)$
  return roadUtils, intUtils, Type_1, Type_2, ActivatedInt

This greedy approach, while intuitive, is limited: it does not incorporate network adjacency into its iterative decision-making. Consequently, the full adjacency utility $U_{ij}^{\text{int}} y_{ij}$is only evaluated ex-post, leading to potential suboptimal solutions. The computation time of this algorithm is roughly $\mathcal{O}(|R|^2 + |R||I|)$. If the number of road $|R|$ is excessively larger than the number of intersections $|I|$, the term $|R|^2$ dominates and thus the computational complexity is $\mathcal{O}(|R|^2)$. If $|I|$ is large enough, then $|R||I|$ dominates, thus the computational complexity is $\mathcal{O}(|R||I|)$.

Heuristic Algorithm That Considers Adjacency

To address the shortcomings of the naive greedy algorithm, we introduce an enhanced heuristic designed to consider network adjacency during the selection process, denoted as Alg^H. This algorithm adheres to a budget-limited iterative structure. Its initial step mirrors the naive approach by selecting the road possessing the highest potential utility. In subsequent iterations, should there be existing selected roads, the algorithm shifts its focus. It identifies a subset of intersections, I'', where exactly one road is already within the `selected_roads` set and the other is a potential candidate. Furthermore, to satisfy a specified degree constraint (limiting a road to d active connections), any intersection pair is excluded if either of

its constituent roads is already associated with d or more established connections. The algorithm then selects the most beneficial road from this refined subset of candidates. Finally, when any road is activated and has a binding maximum degree constraint, we remove all adjacent partnerts of it from candidates.

procedure Alg^H(Roads, Intersections)
  Initialize Numerical Vars: **roadUtils, intUtils**
  Initialize List Vars: **Type_1, Type_2, ActivatedInt, candidates,** selected_roads
  while $B^L > 0$ **do** ▷ $\mathcal{O}(|R|)$
    if selected_roads.empty() **then**
      Run one iteration from Alg^N(Roads, Intersections) ▷ $\mathcal{O}(|R|)$
    else
      Cond1 = Either i or j in selected_roads ∀ (i, j) in Intersections ▷ $\mathcal{O}(|I||R|)$
      Cond2 = Either i or j in candidates ∀ (i, j) in Intersections ▷ $\mathcal{O}(|I||R|)$
      Cond3 = Either i or j's maximum degree constraint is binding ▷ $\mathcal{O}(|I||R|)$
      Int_Subset = Intersections[Cond1 & Cond2 & Cond3] ▷ Define I'' for this iteration
      if Int_Subset.empty() **then**
        selected_roads = [ ] ▷ $\mathcal{O}(1)$
        Continue ▷ $\mathcal{O}(1)$
      else
        i_pairmax, j_pairmax = Int_Subset["U_ij"].idxmax() ▷ $\mathcal{O}(|I|)$
        Use candidates, selected_roads to check whether i or j to select. Assume i. ▷ $\mathcal{O}(|I||R|)$
        Initialize Boolean Var: **built = False**
        if $p_{\text{i_pairmax}}$ >thres **then**
          Update(i_pairmax, Type_2, ActivatedInt, Roads, Intersections) ▷ $\mathcal{O}(1)$
          candidates.remove(i_pairmax) and built = True ▷ $\mathcal{O}(|R|)$
        else
          Update(i_pairmax, Type_1, ActivatedInt, Roads, Intersections) ▷ $\mathcal{O}(1)$
          candidates.remove(i_pairmax) and built = True ▷ $\mathcal{O}(|R|)$
        if built == True and road max_id is binding **then**
          Remove all adjacent partners from IntSubset from candidates ▷ $\mathcal{O}(|I|)$
  return roadUtils, intUtils, Type_1, Type_2, ActivatedInt

The algorithm's integration of adjacency considerations during decision-making is expected to yield a more favorable solution, trading off for a computational complexity of $\mathcal{O}(|R|^2|I| + |R||I| + |R|^2 + |R|) = \mathcal{O}(|R|^2|I|)$.

Experiment Design and Data Generation

Experiment Setting

To ensure both statistical robustness and practical relevance, we conducted experiments using both generated data and real-world data. We designed two distinct experiments: the first measures our algorithms' alignment with the current bike lane policy, while the second focuses on generating recommendations for new policy formulation.

Experiment 1: Comparison with the Current Policy

Initially, we conduct a counterfactual experiment assuming the absence of any constructed bike lanes. We then compare our algorithm's results against the current policy's existing bike lanes using a novel metric: policy similarity Sim_P. This metric is defined as the total length of activated bike lanes in our solution that are already part of the current policy, divided by the total number of roads in our solution's set of activated bike lanes. Let P be the set of existing bike lanes, and let S^{Alg} be the set of proposed bike lanes provided by the algorithm Alg. The policy similarity is defined as:

$$Sim_P(S^{Alg}) = \frac{\sum_{i \in S^{Alg} \cap P} L_i}{\sum_{i \in S^{Alg}} L_i}$$

An elevated policy similarity suggests that the algorithm produces results that are more congruent with prevailing real-world conditions.

Experiment 2: Prospective Policy Formulation

Beyond simply validating our models against existing infrastructure, a key objective was to empower them to inform future bike lane policy. To achieve this, we designed a targeted experiment: we removed all current bike lanes from our dataset, effectively creating a clean slate for optimization. This allowed our models to operate without the constraints of prior construction, simulating a scenario where planners are considering completely new placements. By evaluating and comparing each model's optimization performance under these "greenfield" conditions, we aimed to uncover which inherently suggests a more optimal and effective policy for new bike lane construction, thereby providing valuable insights and inspiration for urban development.

To thoroughly compare our two heuristic algorithms with the Gurobi optimizer, we set up seven different experiments. In each experiment, we systematically changed one specific parameter, which let us closely examine their optimality gaps, the features of their solution maps, and their computational performance under various conditions. Moreover, the seven established scenarios are replicated for the second experiment, Prospective Policy Formulation, with each corresponding index receiving an `_np` suffix. Consequently, a comprehensive set of 14 scenarios is presented in Table 1.

scenarioID	Description	μ	α	B^L	d
1	low budget	0.2	0.2	25,000	3
2	control group	0.2	0.2	50,000	3
3	high budget	0.2	0.2	100,000	3
4	indifferent U_i and U_{ij}	0.5	0.2	50,000	3
5	U_i preferred	0.8	0.2	50,000	3
6	tighter maximum degree constraint	0.2	0.2	50,000	2
7	looser maximum degree constraint	0.2	0.2	50,000	4
1_np	low budget	0.2	0.2	5,000	3
2_np	control group	0.2	0.2	10,000	3
3_np	high budget	0.2	0.2	15,000	3
4_np	indifferent U_i and U_{ij}	0.5	0.2	10,000	3
5_np	U_i preferred	0.8	0.2	10,000	3
6_np	tighter maximum degree constraint	0.2	0.2	10,000	2
7_np	looser maximum degree constraint	0.2	0.2	10,000	4

Table 1: Scenarios with Parameters and Description

Notably, in Experiment 2, the budget B^L was scaled down. This decision accounts for the relatively diminished set of available road candidates, which resulted from the exclusion of pre-existing bike lanes.

Synthetic Data Generation

To enhance the robustness of our algorithm's evaluation, we developed a process for generating synthetic experimental data, complementing our use of real-world datasets. This approach allows us to retain the crucial geometric and adjacency structures from the real-world road network. However, instead of using observed values, we randomly generate normalized demand, length, and danger attributes for each road segment.

When generating synthetic data, the Normal Distribution is a common choice. However, it spans from negative infinity to positive infinity, which poses a problem for our Cobb-Douglas utility function that requires positive normalized demand and length. Therefore, we opt for the Gamma Distribution ($\Gamma(\alpha, \beta)$) as our data generation function, as it is inherently defined only on non-negative real numbers, perfectly suiting our model's requirements. To generate synthetic data, we first perform Gamma distribution estimation on the actual normalized demand, length, and normalized danger datasets. This process yields approximations for each variable's corresponding shape (α [4]) and scale (β) parameters. Subsequently, we leverage the np.random.gamma method to generate the synthetic data. This generation utilizes the previously approximated parameters and matches the length of the original real-world datasets, ensuring structural consistency. The parameter approximation is shown in Fig 2.

[4]Note that the α here is not the parameter we defined in the model. It is the shape parameter of Gamma Distribution.

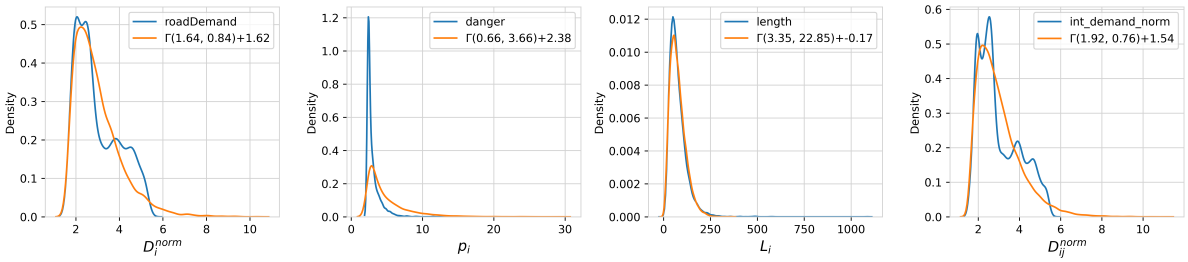


Figure 2: Gamma Approximation for D_i^{norm}, p_i, L_i, D_{ij}^{norm}.

Results and Discussions

Interpretation of Experiment Result Tables

The experiment result of the synthetic data is shown in Table 2, and that of the real world data is shown in Table 3.

Optimality Evaluation: Across most scenarios, Alg^H outperforms Alg^N in terms of objective value, especially when adjacency contributes meaningfully to utility. This is expected, since Alg^H directly incorporates adjacency into its decision process, whereas Alg^N relies only on individual road utility. However, neither heuristic explicitly takes the weight parameter μ into account during the decision process. In high-μ scenarios, where road utility dominates over connectivity, this may weaken the relative advantage of Alg^H, as seen for example in Scenario 5 of the synthetic data experiments.

Computation Time: For all scenarios, Alg^H takes significantly more time to complete optimizing. From scenario 1 to 3, we observe that the computation time incommensurately rises compared to the constant increase in total budget. For scenario 6 where the maximum degree constraint is more aggresive, Alg^H also takes more time to find an intersection that fulfills the constraint. On the other hand, Alg^N is apparently more efficient in terms of computation complexity. While the trade-off between computation time and optimality gap remains large for the heuristic algorithm, it is still economical to prioritize it considering the excessively high computation time for the IP solver when addressing real world data.

Clean-State Scenarios: Both heuristics show higher optimality gaps when we eliminate existing bike lanes, but Alg^H in particular tends to degrade more. One possible explanation is that the removal of existing infrastructure also eliminates many adjacency links in the road network. With a less connected and sparser adjacency matrix, Alg^H loses the structural information on which it relies, reducing its effectiveness. We observe a more aggressive degeneration in the result of real world experiment especially for Alg^H. The visualization of solutions by algorithms is shown in Fig 5. We observe that after eliminating the existing lanes, the result is not informative at all for three solvers. This implies a possible direction of improvement: how do we design a model that considers the adjacency information while not allow any construction on the existing bike lanes?

Policy Similarity: Although both Alg^N and Alg^H feature not promising optimality gap, their solutions are closer to the reality compared to the IP solver in terms of the policy similarity, as defined in the Experiment Setting subsection. This suggests that the two proposed algorithms are both practical and efficient when making decision in reality. The low policy similarity by the IP solver might also suggest that the objective function that the government tried to optimize contains some latent factors that we are unable to access or depict.

	Objective Value (mean)			Optimality Gap				Computation Time (mean)		
				Alg^N		Alg^H				
ID	LR	Alg^N	Alg^H	mean	std	mean	std	LR	Alg^N	Alg^H
1	3724.11	580.59	1246.90	0.84	0.01	0.67	0.01	5.91	0.47	4.94
2	6545.58	1240.84	2428.03	0.81	0.01	0.63	0.01	4.99	1.21	**13.94**
3	10790.32	2854.85	4817.36	0.74	0.01	0.55	0.01	4.70	5.40	**42.90**

4	5161.38	2565.40	2823.52	0.50	0.02	0.45	0.02	5.10	1.11	12.84
5	4835.99	3879.85	3293.48	**0.20**	0.04	0.32	0.03	4.64	1.12	13.17
6	5921.88	1177.81	2022.90	0.80	0.01	0.66	0.01	11.23	3.92	**22.46**
7	6934.88	1265.98	2819.60	0.82	0.01	0.59	0.02	6.04	0.99	12.62
1_np	733.03	112.12	231.16	0.85	0.02	0.68	0.04	1.51	0.06	0.19
2_np	1305.21	218.11	435.47	0.83	0.02	0.67	0.03	1.52	0.11	0.44
3_np	1822.62	345.24	666.50	0.81	0.01	0.63	0.03	1.89	0.23	0.99
4_np	1091.28	514.76	535.19	0.53	0.06	0.51	0.04	1.52	0.11	0.48
5_np	1219.83	834.36	657.88	0.32	0.09	0.46	0.04	1.33	0.11	0.50
6_np	1210.50	231.95	382.56	0.81	0.03	0.68	0.02	1.58	0.12	0.61
7_np	1367.83	226.36	509.64	0.83	0.02	0.63	0.03	1.78	0.16	0.47

Table 2: Result of Experiment based on Synthetic Data

	Objective Value			Optimality Gap		Computation Time			Policy Similarity		
ID	Solver	Alg^N	Alg^H	Alg^N	Alg^H	Alg^N	Alg^H	Solver	Alg^N	Alg^H	Solver
1	3227.62	875.66	1611.44	0.73	0.50	0.68	4.82	115.42	0.675	0.646	0.459
2	5582.36	2156.36	3163.35	0.61	0.43	5.29	18.75	227.44	0.589	0.607	0.471
3	9291.72	4352.92	5563.42	0.53	0.40	27.76	59.47	151.46	0.531	0.486	0.491
4	4883.12	2133.70	3070.19	0.56	0.37	6.99	20.78	190.58	0.589	0.607	0.463
5	4682.41	2111.03	2977.04	0.55	0.36	6.63	21.81	46.56	0.589	0.607	0.455
6	4134.31	1413.09	1954.57	0.66	0.53	8.11	21.31	298.68	0.562	0.501	0.464
7	6850.22	2687.50	3909.58	0.61	0.43	3.57	14.84	480.53	0.625	0.638	0.471
1_np	723.55	72.85	65.96	0.90	**0.91**	0.05	0.06	13.89	-	-	-
2_np	1314.10	287.00	205.20	0.78	**0.84**	0.12	0.22	12.77	-	-	-
3_np	1815.27	486.98	528.06	0.73	**0.71**	0.29	0.73	22.08	-	-	-
4_np	1150.17	318.51	206.99	0.72	**0.82**	0.12	0.21	9.63	-	-	-
5_np	1109.71	350.01	208.79	0.68	**0.81**	0.17	0.20	3.42	-	-	-
6_np	1026.58	195.63	254.54	0.81	**0.75**	0.15	0.46	16.59	-	-	-
7_np	1542.35	313.02	635.51	0.80	**0.59**	0.11	0.49	24.37	-	-	-

Table 3: Result of Experiment based on Real World Data

Visualizations of Solutions

For all experiments, we store the selected indices and their corresponding bike lane level by index, merge back to the original data, and visualize them by levels onto the map. In addition, the bike lanes that have gray slash filled are those that overlap with the currently existing bike lanes. We present several interesting analysis results here.

Comparison Across Algorithms: Consider the baseline: scenario 2, and refer to Fig 3. Based on the (real data) experiment results, our two hand-designed algorithms show better policy similarity than solver. However, the naive algorithm Alg^N suggests a more sparse and disconnected bike lane network. Alg^N provides a more well-connected network, yet still has room to improve. Both algorithms suggest locate most bike lanes in the city center, which is intuitive. In comparison, the solver's suggestion extends to narrower streets and they do not necessarily posit in the central Taipei. In addition, it also generates several star-like isolated bike lane area, which means that the maximum degree constraint with $d = 3$ did not address the problem completely.

Influence of μ: We observe how changing parameter μ affects the optimal solutions induced by the solver. As can be seen in Fig. 4, the structure of the solution changes drastically with different values of μ. For $\mu = 0$, the objective function only considers the intersection utility term, individual road utility (and therefore, components like road dangers) are not taken into account at all. For the other extreme case of $\mu = 1$, only the individual road utility is considered in the objective function, and the resulting cycling network consists only of disconnected type 2 bike-lanes, due to road danger dominating the objective function. For intermediate values, e.g. $\mu = 0.5$, good solutions are obtained, with connected bike lanes, and type 2 bike lanes at key intersections with particularly high danger levels.

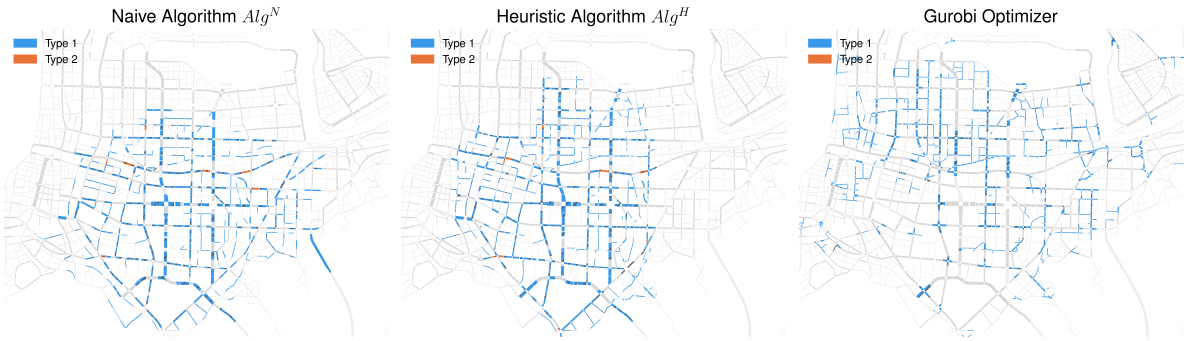


Figure 3: Visualization of Solutions by Algorithms for Scenario 2 (Baseline)

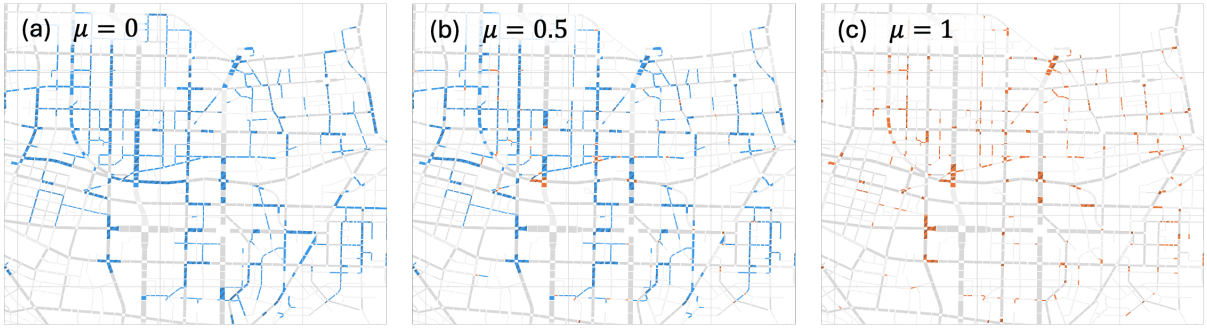


Figure 4: IP optimal solutions for different values of μ. Blue: type 1 bike lanes, Orange: type 2. (shared parameters: $\alpha = 0.2; B_L = 50,000; w = 3$)



Figure 5: Visualization of Solutions by Algorithms for Scenario 2 after Removing All Existing Bike Lanes (Blue: type 1 bike lanes, orange: type 2.)

Conclusion

Our integer program optimizes bike lane networks, balancing demand and continuity in incremental and clean-slate scenarios. Varying μ significantly shifts the final layout: higher μ favors dangerous, disconnected roads, while lower values yield continuous networks. Visual maps illustrate how policy priorities shape the bike lane structure. Heuristic algorithms, notably Alg^H, provide near-optimal solutions rapidly, valuable for quick evaluations and early planning. However, static demand and synthetic costs introduce limitations. Future work should integrate real-time usage data, accurate construction costs, and flexible budgets. Incorporating MRT links and live usage data would better align designs with actual cycling patterns in Taipei.